

# UVM Complete Phase Flow Notes

## UVM COMPLETE PHASE FLOW

### Complete UVM Phase Flow

1. build\_phase
2. connect\_phase
3. end\_of\_elaboration\_phase
4. start\_of\_simulation\_phase

### Runtime Phases

5. pre\_reset\_phase
6. reset\_phase
7. post\_reset\_phase
8. pre\_configure\_phase
9. configure\_phase
10. post\_configure\_phase
11. pre\_main\_phase
12. main\_phase
13. post\_main\_phase
14. pre\_shutdown\_phase
15. shutdown\_phase
16. post\_shutdown\_phase

### Analysis Phases

17. extract\_phase
18. check\_phase
19. report\_phase
20. final\_phase

### Graphical Flow:

build\_phase -> connect\_phase -> end\_of\_elaboration\_phase -> start\_of\_simulation\_phase  
-> pre\_reset\_phase -> reset\_phase -> post\_reset\_phase  
-> pre\_configure\_phase -> configure\_phase -> post\_configure\_phase  
-> pre\_main\_phase -> main\_phase -> post\_main\_phase  
-> pre\_shutdown\_phase -> shutdown\_phase -> post\_shutdown\_phase  
-> extract\_phase -> check\_phase -> report\_phase -> final\_phase

### Purpose Summary:

#### build\_phase:

Create all UVM components.

#### connect\_phase:

Connect TLM ports, exports and sequencer-driver connections.

#### end\_of\_elaboration\_phase:

Verify hierarchy and print topology.

#### start\_of\_simulation\_phase:

Final setup before simulation starts.

pre\_reset\_phase:  
Prepare for reset.

reset\_phase:  
Apply reset to DUT.

post\_reset\_phase:  
Verify DUT initialization after reset.

pre\_configure\_phase:  
Prepare register programming.

configure\_phase:  
Program DUT registers.

post\_configure\_phase:  
Verify configuration.

pre\_main\_phase:  
Prepare traffic generation.

main\_phase:  
Main stimulus execution phase.

post\_main\_phase:  
Drain pending activity.

pre\_shutdown\_phase:  
Prepare DUT shutdown.

shutdown\_phase:  
Stop DUT activity.

post\_shutdown\_phase:  
Cleanup before analysis phases.

extract\_phase:  
Collect coverage, scoreboard and statistics.

check\_phase:  
Determine PASS/FAIL.

report\_phase:  
Print final summary report.

final\_phase:  
Final cleanup and resource release.

UVM 1.1d Flow Used In Router Labs:

build\_phase  
-> connect\_phase  
-> end\_of\_elaboration\_phase  
-> start\_of\_simulation\_phase  
-> run\_phase  
-> extract\_phase  
-> check\_phase

-> report\_phase  
-> final\_phase

Interview Revision Flow:

build\_phase : Create Components  
connect\_phase : Connect Components  
end\_of\_elaboration\_phase : Verify Hierarchy  
start\_of\_simulation\_phase : Final Setup  
run\_phase : Execute Test  
extract\_phase : Collect Results  
check\_phase : Verify PASS/FAIL  
report\_phase : Print Summary  
final\_phase : Cleanup